

Juan Manuel Young Hoyos

SOFTWARE ENGINEER, FRONTEND — REACT & TYPESCRIPT · DATA-HEAVY & REAL-TIME INTERFACES

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Autonomous Air System V&V — Hiring Team

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Application for Software Engineer — Frontend, Autonomous Air System V&V

Dear Helsing Hiring Team,

I am applying for the **Frontend Software Engineer** role on the Autonomous Air System V&V team. Building the interfaces through which engineers understand simulation output and AI behaviour — where the UI is the instrument people use to decide whether a system is safe to fly — is a more interesting problem than most frontend work, and the ethical weight of it is a reason to take it seriously rather than a reason to hesitate.

My work is mostly **React** and **TypeScript** on interfaces that are dense with data. I build monitoring and admin **dashboards** over internal APIs, and as Technical Lead at **Okorum Technologies** I own both the React front end and the services behind it for a national-scale platform, delivered ahead of a hard deadline. I also stood up its **logs, traces and metrics** — so I have been on the other side of this problem, making an opaque distributed system legible to the people operating it.

Frontend performance is something I have measurably done rather than merely believe in: I cut a React/Angular bundle from **3.46 MB to 886 KB**, and I have used Dynatrace to find and fix real user-facing regressions. On real-time and simulation data specifically, I designed a **geohash** spatial algorithm and ran **MQTT** device simulations in Python, feeding a client that displayed results in near real time, and I have worked with **Kafka** streaming.

Your backends are **Rust**, and that is where my own time goes outside work. I run **Project Olivaw** (github.com/Project-Olivaw), a set of robotics libraries in pure Rust: **olivaw-slam**, a 2D lidar **SLAM** implementation with correlative scan matching, keyframing and a log-odds occupancy grid, deliberately built without ROS2 or C++ dependencies; **olivaw-lidar**, an RPLIDAR driver with no C++ SDK or FFI that cross-compiles to aarch64 boards; and **olivaw-cli**, a developer tool that vendors editable Rust components into a project from a curated registry. I visualise live lidar scans with **Rerun** — which is exactly the problem your team has, turning raw sensor and simulation output into something an engineer can read and act on. I have also shipped Go, Python, Node.js and C++ (including a WebAssembly codebase), so navigating an unfamiliar backend to unblock myself is normal for me.

The parts of your day-to-day I am most drawn to are the ones about people: reviewing colleagues' code and RFCs before pushing my own work forward, establishing a shared **component library**, mentoring, and writing things down for asynchronous readers. I have led engineering and QA teams and taught for over a decade. Thank you for your consideration.

Sincerely,

Juan Manuel Young Hoyos

Attached: Resume — Juan Manuel Young Hoyos.pdf